18 Negative Binomial Distributions

Example 18.1. Suppose that 86% of Cal Poly students are CA residents. Randomly select Cal Poly students one at a time, independently[1], until you select 5 CA residents, then stop. Let X be the total number of students selected.

1. Does X have a Binomial distribution? A Geometric distribution? Why or why not?

2. What are the possible values of X? Is X discrete or continuous?

3. Describe in words how you could use simulation to approximate the distribution of X.

4. Compute and interpret $\mathrm{P}(X = 5)$.

5. Compute the probability that the first student is not a CA resident but the next 5 are.

[1]Technically this means *with replacement*, but since the population of Cal Poly students is much larger than the sample we're selecting then independence is a reasonable assumption even if selecting *without* replacement.

6. Why is $P(X = 6)$ different from the probability in the previous part?

7. How many outcomes satisfy the event $\{X = 6\}$? Answer without listing all the possibilities. Hint: careful, it's not $\binom{6}{5}$; why not?

8. Compute and interpret $P(X = 6)$

9. Compute the probability that the first two students are not CA residents but the next 5 are.

10. Why is $P(X = 7)$ different from the probability in the previous part?

11. How many outcomes satisfy the event $\{X = 7\}$? Answer without listing all the possibilities. Hint: careful, it's not $\binom{7}{5}$; why not?

12. Compute and interpret $P(X = 7)$

13. Suggest a formula for the probability mass function of X. Then use the pmf to create a table representing the distribution of X.

14. What seems like a reasonable shortcut formula for $\mathrm{E}(X)$?

15. Use the theoretical pmf to compute $\mathrm{E}(X)$. Did the shortcut formula work?

16. Interpret $\mathrm{E}(X)$ for this example.

17. Compute $\mathrm{Var}(X)$.

18. Would the variance be larger or smaller if we kept selecting until we obtain 10 CA residents instead of 5?

- Perform Bernoulli(p) trials until a fixed number, r, of successes and then stop. Let X count the total number of trials, including the r successes. The distribution of X is defined to be the **Negative Binomial(r, p) distribution**.
- A Negative Binomial distribution is specified by two parameters:
  - r (an integer): the fixed number of successes
  - p (in $[0, 1]$): the fixed marginal probability of success on each Bernoulli trial, or the population proportion of success
- A discrete random variable X has a **Negative Binomial distribution** with parameters r, a positive integer, and $p \in [0, 1]$ if and only if its probability mass function is

$$p_X(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \qquad x = r, r+1, r+2, ...$$

  - In the NegativeBinomial(r, p) situation, there are $\binom{x-1}{x-r} = \binom{x-1}{r-1}$ S/F sequences that result in exactly $x-r$ failures—and thus $r-1$ succeses—in the first $x-1$ trials and success on trial x
  - and each of these sequences has probability $p^r(1-p)^{x-r}$
- If X has a NegativeBinomial(r, p) distribution

$$\mathrm{E}(X) = \frac{r}{p}$$
$$\mathrm{Var}(X) = \frac{r(1-p)}{p^2}$$

Example 18.2. What is another name for a NegativeBinomial(1, p) distribution?

Example 18.3. Xavier bets on red on roulette until he wins 3 bets and then he leaves. After Xavier leaves, Zander arrives and bets on black until he wins 2 bets and then he leaves. Let X be the number of bets that Xavier makes and let Z be the number of bets that Zander makes.

1. Identify the distribution of X and the distribution of Z.

2. Are X and Z independent? Explain.

3. What does $X + Z$ represent in this context? Identify the distribution of $X + Z$ without doing any calculations.

4. Suppose that Yolanda starts making bets on the same game as Xavier, and she bets on black until she wins 5 bets. Does $X + Y$ have a Negative Binomial distribution? Explain.

5. Suppose instead that Zavier bets on the number 7 until he wins 5 bets. Does $X + Z$ have a Negative Binomial distribution in this case? Explain.

- If X and Y are independent, X has a NegativeBinomial(r_X, p) distribution, and Y has a NegativeBinomial(r_Y, p) distribution then $X + Y$ has a NegativeBinomial($r_X + r_Y$, p)
- A NegativeBinomial(1, p) distribution is a Geometric(p) distribution.
- If $X_1, X_2, \dots, X_r$ are independent each with a Geometric(p) distribution, then $X_1 + \cdots + X_r$ has a NegativeBinomial(r, p) distribution.

In a Negative Binomial situation, the number of successes is fixed so what's random is the number of failures and hence the number of trials. Unfortunately, there is not a standard definition of Negative Binomial in terms of what is counted: failures or trials. It's a good idea to always check the documentation of a software package to see what definition is used.

- Perform Bernoulli(p) trials until a fixed number, r, of successes and then stop. Let X count the total number of *failures* (excluding the r successes). The distribution of X is defined to be the **Pascal(r, p) distribution**.

  - The possible values of a NegativeBinomial(r, p) distribution are $r, r+1, r+2, \ldots$
  - The possible values of a Pascal(r, p) distribution are $0, 1, 2, \ldots$

- If X has a NegativeBinomial(r, p) distribution then $(X - r)$ has a Pascal(r, p) distribution.
- Similarly, if Y has a Pascal(r, p) distribution then $(Y + r)$ has a NegativeBinomial(r, p) distribution.
- Some software packages call "Negative Binomial" what we are calling "Pascal".